

Certifying optimized transfer cutoffs

Collatz Lean research project

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Abstract

We give a finite certificate for bounding every transfer request of the classical odd-run proof below a seed threshold. Two residue maxima per possible run length suffice. Lean verifies optimized cutoffs at parameters 61 and $2^{64} - 1$, together with their maximality for this request criterion. All convergence conclusions retain the smaller-transfer hypotheses.

1 Requests and the certificate

Let T be the shortcut Collatz map, $R(n)$ eventual arrival at one, and $A(v)$ the implication $R(3v + 2) \Rightarrow R(27v + 20)$. The bounded-transfer theorem needs only requests satisfying

$$a \geq 2, \quad m > 0 \text{ odd}, \quad 2^a m \leq N, \quad 36v + 27 = 3^a m. \quad (1)$$

If every such v is below U and $A(v)$ holds for all $v < U$, then every positive seed below N reaches one.

Choose L with $N < 2^L$. Any request has $a < L$. For $r \in \{1, 3\}$ define

$$M(B, r) = \begin{cases} r + 4\lfloor (B - r)/4 \rfloor, & r \leq B, \\ 0, & r > B. \end{cases}$$

The certificate checks, for $2 \leq a < L$ and each r ,

$$3^a r \equiv 3 \pmod{4} \implies 3^a M(\lfloor N/2^a \rfloor, r) < 36U + 27. \quad (2)$$

This tests only the largest possible odd part in each eligible residue class. It does not enumerate all seeds or all transfer parameters.

2 Soundness and descent

For a request (1), let $r = m \bmod 4$. Then r is 1 or 3 and $m \leq \lfloor N/2^a \rfloor$. The residue-maximum formula gives $m \leq M(\lfloor N/2^a \rfloor, r)$. Its congruence condition follows from (1). Equation (2) therefore yields $36v + 27 < 36U + 27$, hence $v < U$.

Compose this bound with the exact-request theorem in `BoundedTransfer.lean`. If a target $T^t(27U + 20)$ lies below N , its positivity and the convergence cutoff prove $A(U)$ from the smaller-transfer hypotheses. This is a conditional induction rule, not an assertion that its hypotheses hold for all U .

3 Checked cutoffs and maximality

Lean's ordinary `decide` tactic verifies both certificates:

$$(U, N, L) = (61, 415, 9), \quad (2^{64} - 1, 30786325577727, 45).$$

Their previous sufficient dyadic cutoffs were 128 and 2^{43} respectively. For $U=61$, the next seed 415 has $a = 5, m = 13$ and requests $v = 87 \geq 61$. For the larger U , seed 30786325577727 has $a = 42, m = 7$ and requests $v = 21275914553349625401 \geq U$.

Lean proves that every larger N , with any valid L , fails this request certificate in each case. Maximality concerns this arithmetic request system; it does not rule out other convergence proofs or larger cutoffs obtained by proving additional transfer instances. The formal module `Collatz.RequestCutoff` uses only standard foundations. The Python optimizer proposes the checked outputs; its general correctness and a global descent or covering argument remain unproved. No literature-priority claim is made.